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Pore-Network Model Parameters: Mathematical Definitions

Notation

Symbol	Meaning
R_p	Radius of pore p
R_t	Radius of throat t
$\mathbf{x}_p$	Center coordinate of pore p (from its medial ball)
$\mathbf{x}_t$	Center coordinate of throat t (from its medial ball)
$\mathbf{C}_t = (C_x, C_y, C_z)$	Discrete cross-sectional vector of throat t
$A_t = \lvert \mathbf{C}_t \rvert$	Effective cross-sectional area of throat t
V_p	Volume of pore p
S_p	Surface area of pore p
$l_{pt}^{(1)}, l_{pt}^{(2)}$	Distances from throat center to centers of adjacent pores
$L_{12} = l_{pt}^{(1)} + l_{pt}^{(2)}$	Total center-to-center distance between two connected pores via throat t

Symbol	Meaning
G_t	Shape factor of throat t
G_p	Effective shape factor of pore p
$\mathrm{T}(p)$	Set of throats connected to pore p

Throat Parameters

1. Throat Center and Radius

The **throat center** is defined as the center of the medial ball associated with the throat:

$$\mathbf{x}_t = (\text{tr.mb22()->fk},\ \text{tr.mb22()->fj},\ \text{tr.mb22()->fi})$$

This point lies on the medial axis and represents the geometric "skeleton" location of the narrowest passage between two pores.

The **throat radius** is bounded by the medial ball at the throat and the radii of its two adjacent pores:

$$R_t = \min\left(\max(R_t^{\mathrm{mb}},\ 0.5),\ R_{p_1},\ R_{p_2}\right) + \delta$$

- R_t^{mb}: radius of the maximal inscribed sphere (medial ball) at the throat
- $\delta \sim \mathrm{U}(-0.25,\ +0.25)$: small random perturbation for numerical stability
    - Implemented as:

$$\delta = 0.5 \times \left(0.5 - \frac{\text{rand()}}{\text{RAND\_MAX}}\right)$$

2. Cross-Sectional Vector $\mathbf{C}_t$ and Area A_t

During network construction (`createNewThroats`), for every pair of adjacent voxels belonging to different pores p_1 and p_2, the code detects a face interface and accumulates a signed count into the throat's `CrosArea` vector:

- If the interface is perpendicular to the **x-axis**, then:

$$C_x \leftarrow C_x + (2 \cdot \mathbb{I}[p_2 > p_1] - 1)$$

- Similarly for **y-axis** → C_y, and **z-axis** → C_z.

Here, $\mathbb{I}[\cdot]$ is the indicator function, and the sign ensures symmetry: swapping p_1 and p_2 flips the sign of $\mathbf{C}_t$, but $A_t = /\!/ \mathbf{C}_t /\!/$ remains unchanged.

The **effective cross-sectional area** is then:

$$A_t = /\!/ \mathbf{C}_t /\!/ = \sqrt{C_x^2 + C_y^2 + C_z^2}$$

> **Note**: A_t is dimensionless and proportional to the number of voxel faces connecting the two pores. It serves as a discrete proxy for hydraulic conductance area.

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3. Half-Lengths and Throat Segment Length

Distance from throat center to each adjacent pore center:

$$l_{pt}^{(1)} = \begin{cases} x_t & \text{if } p_1 \text{ is inlet (index 0)} \\ L_x - x_t & \text{if } p_1 \text{ is outlet (index 1)} \\ /\!/ \mathbf{x}_{p_1} - \mathbf{x}_t /\!/ & \text{otherwise} \end{cases}$$

Similarly for $l_{pt}^{(2)}$. Total inter-pore distance:

$$L_{12} = l_{pt}^{(1)} + l_{pt}^{(2)}$$

If $L_{12} < 3$, it is clamped: $L_{12} \leftarrow 3.01$

Each pore is assigned 67% of its half-length (except boundary pores):

$$l_{p_1} = \{\begin{matrix} 1 & \text{if } p_1 \text{ is boundary} \\ 0.67 \cdot l_{pt}^{(1)} & \text{otherwise} \end{matrix}, \quad l_{p_2} = \{\begin{matrix} 1 & \text{if } p_2 \text{ is boundary} \\ 0.67 \cdot l_{pt}^{(2)} & \text{otherwise} \end{matrix}$$

Throat segment length:

$$l_t = \max\left(L_{12} - l_{p_1} - l_{p_2},\ 10^{-7}\right)$$

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4. Throat Shape Factor

Defined as:

$$G_t = \frac{R_t^2}{4A_t}$$

To ensure physical plausibility, bounds are enforced:

$$G_t \leftarrow \begin{cases} \min(0.079,\ G_t/2) & \text{if } G_t \geq 0.09 \\ \max(G_{\text{rand}},\ 0.01) & \text{if } G_t < 0.01 \end{cases}$$

- G_{rand}: small random value (e.g., drawn from uniform distribution)

Pore Parameters

1. Pore Radius

$$R_p = \max(R_p^{\text{mb}},\ 1.0)$$

- R_p^{mb}: radius of the medial ball associated with pore p

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2. Pore Shape Factor

Area-weighted average of connected throat shape factors:

$$G_p = \frac{\sum_{t \in \mathrm{T}(p)} G_t A_t}{\sum_{t \in \mathrm{T}(p)} A_t}$$

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3. Pore Volume Adjustment

An effective cross-sectional area for the pore is estimated as:

$$A_p = \frac{R_p^2}{4G_p}$$

Total weighting area:

$$A_{\text{total}} = \sum_{t \in \mathrm{T}(p)} A_t + A_p$$

Let V_p^{orig} be the original geometric volume of pore p (counted during voxel labeling). Then:

- Adjusted pore volume:

$$V_p = V_p^{\text{orig}} \cdot \frac{A_p}{A_{\text{total}}}$$

- Each connected throat t receives a volume share:

$$V_t \mathrel{+}= V_p^{\text{orig}} \cdot \frac{A_t}{A_{\text{total}}}$$

This ensures total volume conservation while redistributing pore volume based on hydraulic relevance.